

The proposed local-translation characterization is false

Autonomous counterexample packet for arXiv:0901.4115

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Target and answer

Dutkay–Han–Jorgensen ask whether a probability measure μ on $\mathbb{R}^n$ must be spectral if $L^2(\mu)$ carries a strongly continuous unitary group $U(t)$ such that

$$U(t)\chi_{O+t} = \chi_O$$

whenever $O, O+t \subset \text{supp } \mu$ [1].

We propose the following question on a possible characterization of spectral measures. As we will see in section 2.1, the question has a positive answer for atomic measures:

Question. Let μ be a probability measure on $\mathbb{R}^n$. Suppose there is a (strongly continuous) group of unitary transformations $(U(t))_{t \in \mathbb{R}^n}$ on $L^2(\mu)$ such that for every measurable set O , and $t \in \mathbb{R}^n$ with $O, O+t \subset \text{supp}(\mu)$

$$U(t)\chi_{O+t} = \chi_O.$$

Is μ is a spectral measure?

The answer is no. In fact, the hypothesis can be vacuous away from $t = 0$.

Theorem 1. *For every $d \geq 2$, let μ be normalized surface measure on the unit sphere $\mathbb{S}^{d-1} \subset \mathbb{R}^d$. The trivial representation $U(t) = I_{L^2(\mu)}$ satisfies the displayed local-translation condition, but μ is not a spectral measure.*

Proof

The maps $U(t) = I$ form a strongly continuous unitary representation of $(\mathbb{R}^d, +)$. Fix a measurable O and $t \in \mathbb{R}^d$ for which

$$O \subset \mathbb{S}^{d-1}, \quad O+t \subset \mathbb{S}^{d-1}.$$

If $t = 0$, then $U(0)\chi_O = \chi_O$.

Suppose $t \neq 0$. Every $x \in O$ obeys $|x| = |x+t| = 1$, and subtraction of the squared equations gives

$$2x \cdot t + |t|^2 = 0. \tag{1}$$

Consequently,

$$O \subset \mathbb{S}^{d-1} \cap \{x : x \cdot t = -|t|^2/2\}.$$

This intersection is empty, a point, or a $(d-2)$ -sphere. In every case it has $(d-1)$ -dimensional surface measure zero. Thus $\mu(O) = 0$. Applying the same argument to $O+t$ and the translation $-t$ gives $\mu(O+t) = 0$. Hence $\chi_O = \chi_{O+t} = 0$ as elements of $L^2(\mu)$, and

$$U(t)\chi_{O+t} = 0 = \chi_O.$$

The required condition holds for every O and t .

It remains to rule out spectrality. Iosevich–Lai–Liu–Wyman prove that surface measure on the smooth boundary of a convex body with everywhere positive Gaussian curvature admits no Fourier frame [2, Theorem 1.1]. The unit sphere is such a boundary. Multiplying a measure by a positive constant does not affect whether an exponential family is a frame. If μ had an orthonormal basis of exponentials, that basis would in particular be a Fourier frame (with frame bounds 1, 1), contradicting their theorem. Therefore μ is not spectral. $\square$

A general obstruction

The same argument isolates the defect in the proposed condition. If a probability measure ν with support K satisfies

$$\nu(K \cap (K + t)) = 0 \quad (t \neq 0), \quad (2)$$

then the trivial unitary group satisfies the question's hypothesis. Thus that hypothesis cannot characterize spectrality on any class containing a non-spectral measure with property (2). Curved hypersurface measure gives a particularly concrete family: intersections with nontrivial translates have codimension at least one.

This counterexample even survives the stronger, pointwise-looking formulation $U(t)f(x) = f(x + t)$ for μ -almost every x for which both points lie in the support: for $t \neq 0$, that region is still μ -null.

Proof intuition

The local-translation identity only tests portions of the support that remain inside the support after translation. A curved surface has essentially no such portions: it meets a distinct translate only in lower dimension. The identity therefore contains no information, and even the identity representation passes it. Positive curvature simultaneously supplies a known obstruction to any Fourier frame, turning this vacuity into a full counterexample.

References

- [1] D. E. Dutkay, D. Han, P. E. T. Jorgensen, *Orthogonal exponentials, translations, and Bohr completions*, arXiv:0901.4115.
- [2] A. Iosevich, C.-K. Lai, B. Liu, E. Wyman, *Fourier frames for surface-carried measures*, arXiv:1905.07032.